Risk Factors and Pre-operative Stratification for Intraoperative Hypotension

ASA Physical Status, MPOG Collaborative & Clinical Predictors

1. Pre-operative Patient Factors

The following patient-level factors are independently associated with IOH:

AGE: Patients ≥65 years are at significantly elevated risk due to reduced baroreceptor sensitivity, decreased cardiac reserve, and impaired vascular compliance. Risk increases linearly beyond 70 years.

PRE-EXISTING HYPERTENSION: Paradoxically increases IOH risk. Hypertensive patients require higher perfusion pressures; anesthetic vasodilation produces disproportionate MAP drops. Up to 40% of hypertensive patients develop IOH during general anesthesia.

DIABETES MELLITUS: Autonomic neuropathy from chronic diabetes impairs the sympathetic reflex response to hypotension, blunting compensatory tachycardia and vasoconstriction.

CARDIAC DISEASE: Reduced ejection fraction (<50%), valvular disease (aortic stenosis), and ischemic heart disease increase vulnerability to anesthetic-induced myocardial depression.

ASA PHYSICAL STATUS: ASA ≥3 correlates with IOH incidence:

- ASA I: ~8% IOH rate
- ASA II: ~18% IOH rate
- ASA III: ~35% IOH rate
- ASA IV: ~52% IOH rate

2. Anesthetic Technique Factors

GENERAL ANESTHESIA: Propofol causes significant vasodilation and negative inotropy; sevoflurane and isoflurane reduce SVR; high-dose opioids can cause vagally-mediated bradycardia. Induction is the highest-risk period.

SPINAL ANESTHESIA: Sympathectomy below the block level causes dramatic SVR reduction. High spinals (T4 level) additionally impair cardiac sympathetic innervation, removing compensatory tachycardia. Incidence of hypotension with spinal: 30–60%.

EPIDURAL ANESTHESIA: More gradual onset than spinal, but still produces sympathetic blockade and vasodilation; risk increases with high block levels.

COMBINED GENERAL + NEURAXIAL: Additive vasodilatory effects; highest-risk combination.

3. Surgical Factors

PROCEDURE TYPE: Major abdominal (especially bowel resection, hepatobiliary) and thoracic procedures involve significant evaporative losses and third-space fluid shifts.

POSITIONING: Steep Trendelenburg position (robotic procedures) causes redistribution; lateral decubitus reduces venous return from dependent limbs.

DURATION: Prolonged surgery (>3 hours) increases cumulative anesthetic drug exposure and fluid shifts; IOH incidence rises linearly with case duration.

BLOOD LOSS: Expected hemorrhage >500 mL is a major independent predictor.